ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

FINDING GOOD REPRESENTATIONS OF DATA

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Uncovering Latent Structure

We have seen various models of **latent structure**, e.g.

- clusters
- linear subspaces

and algorithms that map data to a representation in which this structure is exposed.

Uses of the latent representation:

- As a way to better understand the data
- As a representation for subsequent learning or processing

Embedding Data into Simpler Spaces

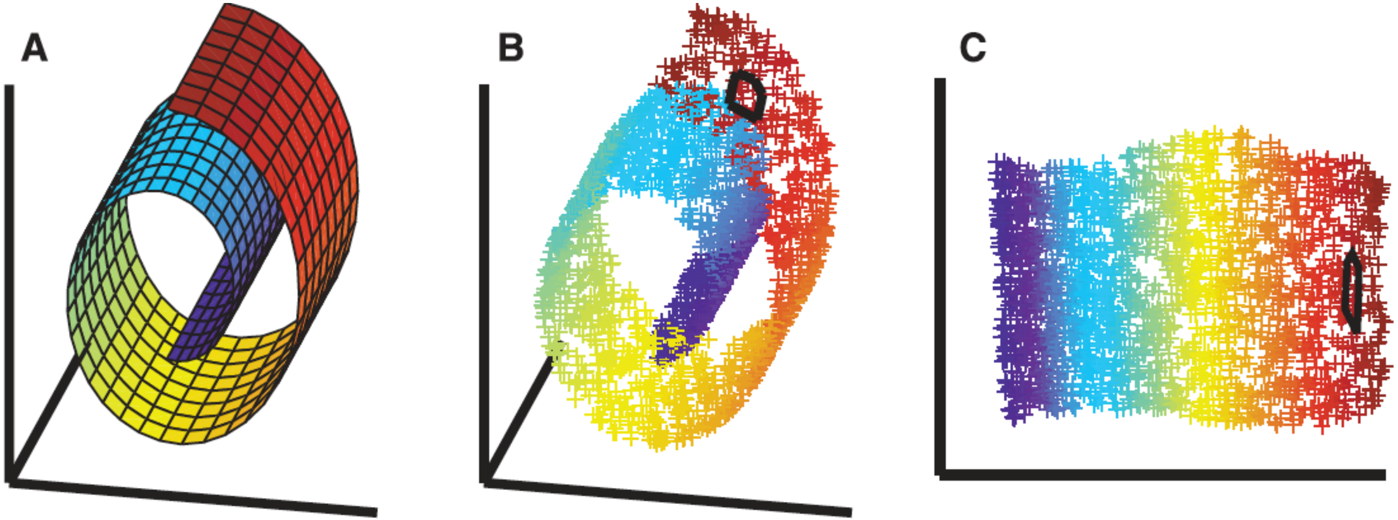

Figure: S. Roweis and L. Saul (2000). Nonlinear dimensionality reduction by locally linear embedding.

- **Data space:** weird geometry, weird distance/similarity structure
- **Ideal embedding:** into a space where familiar geometry (e.g., Euclidean) and simple methods (e.g., linear separators) work well